MIL-M-60557G (AR)
23 NOVEMBER 1987
SUPERSEDING
MIL-M-60557F (AR)
23 September 1983

MILITARY SPECIFICATION

MICRO-REPRODUCTION SYSTEM FOR ENGINEERING DOCUMENTS AND ENGINEERING RELEASE RECORDS

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the procedure and requirements for the microfilming of engineering documents and Engineering Release Records (ERRs), key punching, verifying, interpreting, sequencing aperture and tabulating cards, and mounting microfilm images into aperture cards for the Engineering Data Micro-Reproduction System.

1.2 Definitions. For definitions of words and phrases used in this specification, see 6.3.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-9868	Microfilming of Engineering Documents, 35mm, Requirements for.
MIL-C-9877	Cards, Aperture.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-ESC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC A4240 EDRS

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MIL-M-38761	Microfilming and Photographing of Engineering/Technical Data and Related Documents: PCAM Card Preparation, Engineering Data Micro-reproduction System, General Requirements for, Preparation of.

STANDARDS

MILITARY

MIL-STD-804	Formats and Coding of Tabulating Aperture Cards for Engineering Data Micro-reproduction Systems.

2.1.2 Other Government publications. The following other Government publiciations form a part of this specification to the extent specified herein.

FORMS AND CATALOGS

DD Form 1562	Dual Purpose Engineering Document Card (Card Code H/T).
H4/H8	Cataloging Handbook - Commercial and Government Entity.

(Copies of specifications, standards, handbooks, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. Each document in the ERR package shall be microfilmed, one sheet per frame, five images each, as specified herein. A set of Dual Purpose engineering document cards, DD Form 1562, card code T (lower legend), shall be prepared in accordance with 3.6 for each document to be microfilmed.

3.2 Preproduction sample. A minimum of 500 finished tabulating nonaperture and 500 aperture cards from the first roll of film processed shall be furnished for Government inspection and approval prior to the start of full production.

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3.3 Class of microfilm. Microfilm shall be Type 1, Class 1, silver hallide Camera microfilm (negative-type) with clear line image. Reduction ratios used shall be in accordance with MIL-M-9868.

3.4 Class and format of aperture cards. Aperture cards shall be Type 1, Cold Seal, Pressure Sensitive Microfilm Carrier, Class 1, Microfilm Carrier on Face, in accordance with MIL-C-9877. Card format, DD Form 1562, per MIL-STD-804, shall be surface-printed in black ink on the face of the cards.

3.5 Mounting. Mounting requirements set forth in MIL-M-38761 shall apply. Each frame of microfilm, except erroneously repeated, defaced, punched out, and illegible frames, shall be mounted in a properly identified aperture card. Targets shall be mounted on unpunched aperture cards and submitted to the Government for resolution verification to ensure compliance with MIL-M-9868.

3.6 Sets of aperture cards. (See 6.2.) The dual purpose engineering document cards, DD Form 1562, card code T, (lower legend) that are the end product shall be furnished in discrete sets as follows:

a. One card, plain, without color striping, nonaperture.

b. One card, plain white, aperture.

c. One card, solid blue, aperture.

d. One card, solid green, aperture.

e. One card, with 1/4 inch brown stripe printed across the face above the 12 punch position, aperture.

f. One card, solid yellow, aperture.

3.7 Illegible frames of microfilm. Erroneously repeated, punched out, or illegible frames of microfilm shall be removed from the roll during the mounting operation. The contractor shall assure the replacement of all illegible frames of microfilm.

3.8 Card punching and verifying. A set of cards (3.1) shall be punched for each set of duplicate frames or images of microfilm in the roll. Extracting of data onto alignment sheets for card punching and verifying shall be made from the original document. Each set of cards shall be punched in accordance with requirements of appendix A and as clarified below. Data fields not included in appendix A shall not be punched. Punched nonaperture cards of each set shall be verified, and the remainder of the cards in the sets shall be reproduced from the verified, nonaperture cards.

NOTE: Sheets larger than 34 inches by 44 inches shall be microfilmed using more than one frame or camera exposure. A separate card with a unique sheet/frame identification shall be punched for each frame of a multi-frame sheet.

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3.8.1 Data fields. Detailed information for data fields is as follows:

3.8.1.1 Type of document (columns 1-2, full field). Prefix codes for types of documents shall be in accordance with appendix B.

3.8.1.2 Document numbers (columns 3-17, left justified). Enter the identifying number of the document, closing empty spaces:

EXAMPLES:

Ordnance drawing number	7 6 5 3 4 2 1
Ord engr stds dwg BFAX number	B F A X 1 0 . 7
MS drawing number	M S 5 0 0 8 2
List	7 6 5 3 4 2 1
ERR or Notice of Revision (NOR)	F 7 J 2 0 0 1

3.8.1.3 Contractor and Government Entity (CAGE) (columns 18-22, full field). Enter the CAGE code (formerly Federal Supply Code for Manufacturers (FSCM) of the government design agency responsible for approving the ERR and the NOR. When logging a document within the ERR package, use the CAGE code of the original government design agency for the drawing or associated document.

3.8.1.4 Sheet number (columns 23-25, right justified - zero filled). The sheet number of a multi-sheet document shall be entered in this field. The ERR, AMC Form 1724, will be entered as sheet 1 and the total number of sheets comprising the ERR and NOR (not all documents) will be entered in columns 28-30. (See 3.8.1.6.) The remaining documents comprising the package (for example, AMC Form 1724A, ERR continuation sheet(s) (if applicable), first NOR, second NOR) will be entered sequentially in ascending numerical order.

3.8.1.5 Revision (columns 26-27, right justified - zero filled). The revision letter or number shall be entered in this field. Leave blank when documentation type is an ERR or a NOR. Field is also left blank for initial release documents which have the symbol "-" in the revision block.

3.8.1.6 Number of sheets (columns 28-30, right justified - zero filled). The total number of sheets of a document shall be entered in this field for sheet 1 only.

3.8.1.7 Frame number (columns 31-32, right justified - zero filled). Enter the frame number of each frame in a series of frames. A single frame shall be frame number 1.

3.8.1.8 Number of frames (columns 33-34, right justified - zero filled). Enter the number of frames in the series. A single frame shall be frame number 1.

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3.8.1.9 Document date (columns 35-40, full field). Enter the date of the document as a six-digit number. The first two digits represent the month. The digit for the first nine months will be preceded by a zero. The second two digits represent the day of the month. The first nine days of the months will be preceded by a zero. The last two digits will be the last two numerals of the year. The last digit of each portion of the date shall be entered in the right column and nonsignificant zeros shall be used to fill out the field. If the document being microfilmed is an ERR, enter the date prepared. If the document is a NOR, enter the date appearing in block 10 of DD Form 1695.

3.8.1.10 ERR number (columns 41-46, full field). Enter the last six characters of the identification number assigned to the ERR.

3.8.1.11 Rights (column 47, full field). Enter the rights status of the information of the document. The rights status codes are as follows:

a. U-Code. "U" signifies that the Government has an unlimited right to use the document so coded.

b. L-Code. "L" signifies that the Government has only a limited right to use the document so coded.

3.8.1.12 Control activity (columns 48-49, full field). Enter the code "DM."

3.8.1.13 Drawing size (column 50, full field). The drawing size letter shown on the drawing shall be entered in this field. Documents such as ERRs, NORs, QAPs, and associated lists, which are 8-1/2 inches x 11 inches will be encoded as "A." Computer prepared parts lists will be encoded as "B."

3.8.1.14 Card code (column 51, full field). Code T shall be entered in this field.

3.8.1.15 Security classification (column 52, full field). The security classification of the microfilm image in an aperture or of the document identified in a tabulating (nonaperture) card, shall be entered in this field, utilizing the following codes:

N - Unclassified
C - Confidential
M - Confidential - Modified Handling Authorized
S - Secret
T - Top Secret
E - Confidential Restricted Data
F - Secret Restricted Date
G - Top Secret Restricted Data
H - Confidential - Formerly Restricted Data
J - Secret - Formerly Restricted Data
K - Top Secret - Formerly Restricted Data

NOTE: Aperture cards mounted with images of classified documents must be clearly marked for identification as specified in 3.9, below.

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3.8.1.16 Revised code (column 53, filled or blank). Enter a "-" (11 punch) if document is backfiled, superseded, or obsolete.

3.8.1.17 Columns 54-76. Leave blank.

3.8.1.18 ERR number - 1st character entry (column 77, full field). Enter the first character of the identification number assigned to the ERR.

3.8.1.19 Columns 78-79. Leave blank.

3.8.1.20 Deck code (column 80, full field). Punch the deck codes to identify the properly segregate cards into decks as follows:

CODE	
0	Plain, without aperture
1	Plain, white with aperture
2	Solid blue with aperture
3	Solid green with aperture
4	Brown stripe with aperture
5	Solid yellow with aperture

3.8.2 Interpretation. Information punched into data fields shall be interpreted above the 12 punch location in solid line characters at least .075 inches high, in accordance with interpreter bar assignments shown in appendix A. All aperture cards shall be interpreted prior to mounting the microfilm.

3.9 Identification of cards containing classified information. Cards containing classified information shall meet the current requirements of MIL-STD-804 for security markings.

NOTE: Classified images shall be mounted in special aperture card stock and identified clearly.

3.10 Sequencing of cards. Completed cards shall be sequenced as follows:

Sorting sequence no.	Field	Columns From	Columns To
1	Deck code	80	80
2	Frame number	31	32
3	Sheet number	23	25
4	Type document code	1	2
5	Document number	3	17

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4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance provisions shall be in accordance with MIL-M-38761.

5. PACKAGING

5.1 Preservation, packing, and packaging. Preservation, packing, and packaging shall be level C in accordance with MIL-M-38761.

6. NOTES

6.1 Intended use. The end item documentation and related services are intended for use in the Department of the Army Armament, Munitions and Chemical Command Engineering Data Micro-reproduction System, which includes use of automated microfilm storage and retrieval equipment.

6.2 Data requirements list and cross-reference. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the date requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-part 27.410-6 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraph.

Paragraph no.	Data requirement title	Applicable DID no.
3.1, 3.2, 3.4, 3.5, 3.8, 3.9, and 3.10	Aperture Cards (Engineering Documents and Release Records)	DI-EDRS-80460

(Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Definitions. The following definitions apply to words and phrases used in this specification and to the specifications and standards listed in 2.1. Additional definitions to micrographic words and phrases are contained in NMA (National Micrographic Association) Standard MS-100-1971.

6.3.1 Aperture card. An aperture card is a tabulating card with a rectangular hole specifically prepared for the mounting of a frame of microfilm.

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6.3.2 Camera card. A camera card is a tabulating card containing unexposed and unprocessed microfilm which will be exposed and processed for the express purpose of producing or creating a micro-image from an original document.

6.3.3 Copy card. A copy card is a tabulating card with a frame of unexposed and unprocessed microfilm mounted in or over a rectangular hole for subsequent exposure and development of the microfilm which is mounted in the card.

6.3.4 Data field. A data field consists of one or more columns on an aperture, copy, or tabulating card reserved for specific information entered in a specified manner.

6.3.5 EAM card. An EAM card is an Electronic Accounting Machine card.

6.3.6 Engineering Data Micro-Reproduction System (EDMS). The EDMS is a DOD program established to standardize and implement a common micro-reproduction system to be used throughout the Army, Navy, Air Force, Defense Supply Agency, and industrial suppliers to DOD in the preparation of 16 and 35mm microfilm, 105mm and microfiche film copies of engineering/technical data and related documents and the preparation of PCAM (aperture, camera, copy, and tabulating) cards.

6.3.7 Nonaperture card. A nonaperture card is a tabulating card without a rectangular hole specifically prepared for the mounting of the frame of film.

6.3.8 PCAM card. A PCAM card is a Punch Card Accounting Machine card.

6.3.9 Tabulating card. A tabulating card is a PCAM (EAM) card on which data are entered by use of punched holes or other means that can be sensed by a machine so that it can sort, collate, list, total, or otherwise manipulate the card or the data.

6.4 Subject term (key word) listing.

Aperture card
Card punch
Data fields
Engineering Release Record (ERR)
Microfilm
Notice of Revision (NOR)

6.5 Changes from previous issues. Asterisks or vertical lines are not used in this revision to identify changes with respect to previous issues due to the extensiveness of the changes.

6.6 Previous issues. MIL-M-60557F(AR) and associated amendments were for U.S. Army Armament Research, Development and Engineering Center use only, and were not forwarded to the Naval Publications and Forms Center for publication.

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APPENDIX A

CARD PUNCHING AND INTERPRETING REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix provides guidance for dual purpose engineering document cards (code T) DD Form 1562 per MIL-T-804, and is a mandatory part of this specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS.

This section is not applicable to this appendix.

30. DESCRIPTION AND REQUIREMENTS

Data description	Card punching columns	Interpreting 60-BAR	Interpreting 80-BAR
Type of document code	01-02	01-02	02-03
Document number	03-17	04-18	06-20
CAGE (code identification)	18-22	54-58	73-77
Sheet number	23-25	21-23	29-31
Revision level	26-27	19-20	26-27
*Number of sheets	28-30	25-27	34-36
Frame number	31-32	29-30	39-40
Number of frames	33-34	31-32	42-43
Document date - Month	35-36	41-42	53-54
Day	37-38	43-44	55-56
Year	39-40	48-49	57-58
ERR number (last 6 characters)	41-46	34-39	46-51
Rights	47	50	67
Control activity	48-49	51-52	68-69
Document size	50	46	61
Card code T	51	-	-
Security classification	52	60	80
Revised code	53	-	-
ERR number (first character)	77	33	45
Deck code	80	-	-

*Number of sheets is punched and interpreted only for sheet 001.

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APPENDIX A

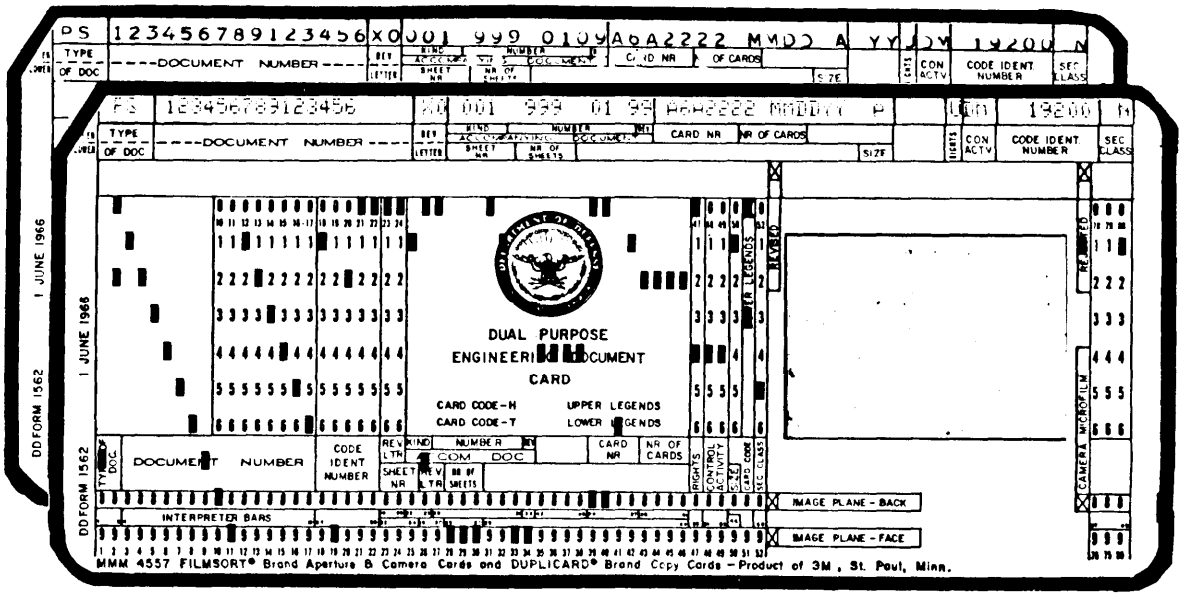


Figure 1. Interpreting example.

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APPENDIX B

APPROVED PREFIXES

10. SCOPE

10.1 Scope. This appendix provides guidance for the types of document codes to be used for dual purpose document cards (code T) DD Form 1562, per MIL-T-804, and is a mandatory part of this specification. The information contained herein is intended for compliance.

20. APPLICABLE DOCUMENTS.

This section is not applicable to this appendix.

30. TYPES OF DOCUMENT CODES

Prefix	Description or title
Blank	Drawings
LQ	List of SQAPs
ED	List of equipment - Depot installed
EM	List of equipment - Manufacturer installed
ET	List of equipment - Troop installed
SQ	Quality Assurance Provision (QAP)
LL	*EPL - Listing of parts list
LC	*EPL - Listing of contents
LD	*EPL - List of drawings
LS	*EPL - List of specifications
LP	*EPL - List of parts
GL	*Gage list
IL	Index list
LI	*Index of inspection equipment lists
EL	List of inspection equipment
LN	*List of inspection numbers
DL	Data list
PL	Parts list
PS	Packaging data sheets
IE	List of index equipment numbers (combined list)
RR	Engineering Release Record (ERR)
NO	Notice of Revision (NOR)

*Old prefix - Document still in use.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-28557G	MICRO REPRODUCTION SYSTEMS FOR ENGINEERING DOCUMENTS

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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